PPL degradation (%) relative to the original GPT-2 (Model PPL) / (Original PPL)

Dataset	Steps(Number of merged pairs)	LinearMultiply (If you test)	Remove	Average	NonlinearMultiply
DBpedia	1	68.7%	4.0%	69.9%	4.1%
DBpedia	2	117.6%	19.4%	184.9%	14.6%
DBpedia	3	250.1%	41.4%	395.2%	31.6%
DBpedia	4	516.9%	86.0%	696.4%	61.5%
DBpedia	5	850.2%	174.8%	1271.1%	93.0%
AG News	1	86.9%	2.9%	71.1%	2.4%
AG News	2	153.8%	21.2%	182.8%	22.0%
AG News	3	308.1%	50.9%	381.5%	40.8%
AG News	4	662.2%	101.5%	635.5%	85.3%
AG News	5	1083.3%	191.4%	998.5%	123.8%
IMDb	1	42.5%	3.4%	47.9%	3.0%
IMDb	2	89.7%	14.9%	127.5%	12.8%
IMDb	3	182.5%	40.3%	258.1%	25.9%
IMDb	4	392.1%	76.8%	440.9%	49.5%
IMDb	5	658.0%	128.8%	735.6%	77.5%

Proposed model details

Steps(Number of merged pairs)	Merge plan	Parameter reduction	DBpedia PPL	AG News PPL	IMDb
Original	—	0.00%	30.99	51.82	37.20
1	1-2	5.70%	32.27	53.04	38.31
2	1-2;4-5	11.39%	35.53	63.19	41.97
3	1-2;4-5;8-9	17.09%	40.78	72.98	46.84
4	1-2;4-5;6-7;8-9	22.78%	50.07	96.04	55.63
5	1-2;4-5;6-7;8-9;10-11	28.48%	59.81	115.97	66.03